

PSP Yearly partitioning for Monolith & Audit database

Audit database

Partitioning scripts : [git](#)

- Ensure there are no data/rows in overflow partitions

```
select count(*) from psp_source_system_transmission_9999;
```

- If there is data, then move data from overflow partition to respective year/month partitions and delete from overflow partitions. Below is an example from SYS Audit database.

```
insert into ibobadm.psp_source_system_transmission_m012023 select *
  from ibobadm.psp_source_system_transmission_9999_from_qbdt
  where created_date between to_date('2023-01-01', 'yyyy-mm-dd') and to_date('2023-01-31', 'yyyy-mm-dd');

delete from ibobadm.psp_source_system_transmission_9999_from_qbdt
where created_date between to_date('2023-01-01', 'yyyy-mm-dd') and to_date('2023-01-31', 'yyyy-mm-dd');
```

- Detach Overflow partition from Main table

```
alter table ibobadm.psp_source_system_transmission detach partition psp_source_system_transmission_9999;
```

- check for invalid indexes. There should not be any.

```
SELECT * FROM pg_class, pg_index WHERE pg_index.indisvalid = false AND pg_index.indexrelid = pg_class.oid;
```

- create partitions for next year
 - [2023](#)
 - [2024](#)
- Validate newly created partition. There should be 12 partitions for months and each month will have 7 partition

```
select substring(tablename, 35, 4), substring(tablename, 32, 3), count(*)
from pg_tables
where schemaname = 'ibobadm' and tablename like 'psp_source_system_transmission_%_from_%'
group by substring(tablename, 35, 4), substring(tablename, 32, 3)
order by substring(tablename, 35, 4), substring(tablename, 32, 3);
```

- Ensure all sub-partitions has PK. Check for each year sub-partition created in above step.

```

select tab.table_schema,
       tab.table_name,
       tco.constraint_name,
       string_agg(kcu.column_name, ', ' ) as key_columns
from information_schema.tables tab
     left join information_schema.table_constraints tco
           on tco.table_schema = tab.table_schema
             and tco.table_name = tab.table_name
             and tco.constraint_type = 'PRIMARY KEY'
     left join information_schema.key_column_usage kcu
           on kcu.constraint_name = tco.constraint_name
             and kcu.constraint_schema = tco.constraint_schema
             and kcu.constraint_name = tco.constraint_name
where tab.table_schema = 'ibobadm'
     and tab.table_name like 'psp_source_system_transmission_%2024_from_%'
group by tab.table_schema,
         tab.table_name,
         tco.constraint_name
order by tab.table_schema,
         tab.table_name;

```

- Ensure all newly created sub-partitions has check constraints. There should be **3** check constraint on each partition

```

select tab.table_schema,
       tab.table_name,
       count(*)
from information_schema.tables tab
     left join information_schema.table_constraints tco
           on tco.table_schema = tab.table_schema
             and tco.table_name = tab.table_name
             and tco.constraint_type = 'CHECK'
             and tco.constraint_name not like '%null%'
     left join information_schema.key_column_usage kcu
           on kcu.constraint_name = tco.constraint_name
             and kcu.constraint_schema = tco.constraint_schema
             and kcu.constraint_name = tco.constraint_name
where tab.table_schema = 'ibobadm'
     and tab.table_name like 'psp_source_system_transmission_%2024_from_%'
group by tab.table_schema,
         tab.table_name
order by tab.table_schema,
         tab.table_name;

```

- Attach overflow partition back to main table by modifying it with new date/time range

```

alter table ibobadm.psp_source_system_transmission attach partition psp_source_system_transmission_9999 FOR
VALUES FROM ('2025-01-01 00:00:00') TO (MAXVALUE);

```

- Check for invalid indexes. There should not be any

```

SELECT * FROM pg_class, pg_index WHERE pg_index.indisvalid = false AND pg_index.indexrelid = pg_class.oid;

```